

Python Mini Projects

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Preface

Welcome to the **Python Mini Projects** collection!

This book serves as a comprehensive guide and repository for a variety of easy-to-understand Python small projects. Whether you are a beginner looking to improve your programming skills or an expert seeking a playground to share knowledge, this collection has something for you.

Aim of the Project

As a Python newbie, I understand the problems that people face when they first begin studying and attempting to understand various Data Science concepts, particularly Python. This project is designed for folks who are just getting started with Python principles and exploring GitHub as “contributors.”

My goal is to build a common playground where everyone can learn and share knowledge.

Let’s “folk-ing” create amazing things together!

1 Introduction

This book documents a collection of Python mini-projects. Each project is designed to be simple, educational, and easy to run.

1.1 How to use this book

You can browse the [Projects list](#) to find a project that interests you. Each project is located in its own directory in the GitHub repository.

1.2 Contributing

We welcome contributions from everyone! If you have a simple Python project you'd like to share:

1. **Fork** the repository.
2. **Clone** your fork.
3. **Create a folder** for your project.
4. **Add your code** and a `README.md`.
5. **Submit a Pull Request**.

For more detailed instructions, please refer to the [repository README](#).

2 Projects

This section lists all the Python mini-projects available in this repository.

2.1 Project List

Project Name	Description
Address Validator	Validates email or physical addresses.
Animalese_translator	Translates text into Animal Crossing style sounds.
Automated_Mailing	Automates the process of sending emails.
Binary_Search_Tree	Implementation of a Binary Search Tree data structure.
Binary_tree	General Binary Tree implementation.
Caesar_Cipher	Classic Caesar cipher encryption and decryption.
Cat_command	A Python implementation of the Unix <code>cat</code> command.
Caterpillar_Game	A simple snake-like game.
Chess_Game	A full chess game implementation.
Chinese_FlashCard	Flashcard app for learning Chinese.
Clip_Organizer	Tool to organize clipboard content.
Color_Game	A game where you guess or match colors.
Connect-Four	The classic Connect-Four strategy game.
Converter	General purpose unit converter.
Converting_Roman_to_Integer	Tool to convert Roman numerals to integers.
Convoys_GameofLife	Implementation of Conway's Game of Life.
Crud_in_flask	A simple CRUD application using the Flask web framework.
Currency_Converter	Converts between different world currencies.
Demerge_pdfs	Splits a single PDF into multiple files.
Dice_Rolling_Stimulator	Simulates rolling one or more dice.
Dictionary	A simple digital dictionary.

Project Name	Description
Diff_Utility	Compares two files and shows the differences.
Download Audio	Tool to download audio from various sources.
Egg_Catcher	A fun game about catching falling eggs.
Email Slicer	Extracts the username and domain from an email.
Encode_Morse.py	Encodes text into Morse code.
Execute Shell Command	Run shell commands directly from Python.
Exercise-Timer	A timer specifically designed for workout routines.
Expense_Tracker	Helps track daily expenses and budget.
ExtractPhoneNumberEmail	Extracts contact info from text files.
Face_Recognition	Basic facial recognition using Python libraries.
Fancy_Text_Generator	Converts plain text into stylized “fancy” text.
Finance_Tracker	More advanced tracking for personal finances.
Finding_Lanes	Computer vision project to find road lanes.
Firebase_Authentication_Using_Flask	Flask app with Firebase auth integration.
Geo_Plot_Using_Folium	Visualizes geographical data on interactive maps.
GK_Maestro	A general knowledge quiz game.
Google_Translate	Interface for Google Translate API.
HangMan	The classic word guessing game.
Hangman_Game	Another version of the Hangman game.
Hashed_and_Salted_Pass	Demonstrates secure password hashing.
image_comparator	Compares two images for similarities.
Image_compressor	Reduces image file size while maintaining quality.
images_to_pdf_converter	Converts a series of images into a PDF document.
img_to_ascii	Converts images into ASCII art.
infix_postfix_calculator	Converts and evaluates mathematical expressions.
Investment Calculator	Calculates potential returns on investments.
IP_Locator	Finds the geographical location of an IP address.
Lazy_Pong	A simple version of the classic Pong game.
linked_lists	Demonstrates linked list data structures.

Project Name	Description
lorem_in_python	Generates Lorem Ipsum placeholder text.
Madlibs	A lighthearted word-replacement game.
Mail_Checker	Validates if an email address exists or is reachable.
Matchmaker	A fun game to find matches between people.
Math_Game	Educational game focused on math problems.
Minesweeper_game	The classic Minesweeper puzzle game.
minionGame	A word-game project based on “The Minion Game”.
mnist_digit_recognizer	Machine learning model to recognize handwritten digits.
Morse_code_beep	Plays Morse code as audible beeps.
Motion_Detection	Detects motion through a webcam feed.
Music-Player	A basic application to play audio files.
NASA_Image_Extraction	Fetches and displays images from NASA’s API.
Number Guessing	Guess the randomly selected number.
Number Guessing Upper Boundary	Number guessing with configurable difficulty.
Object_Detection	Uses AI to identify objects in images or video.
Othello-Reversi-Game	Implementation of the Othello/Reversi board game.
Password Generator	Creates secure and random passwords.
Password_Generator_2	Another version of password generation.
Password_Manager	Stores and manages passwords securely.
Plotter	Tool to create various data visualizations.
Port Scanner	Checks for open ports on a network host.
PostgreSQL_Dumper	Backs up PostgreSQL databases.
Prefix_Trie	Data structure for efficient string searching.
QR Code Generator	Generates QR codes from text or URLs.
qr_with_logo	Generates QR codes with embedded logos.
Regression using ANN	Predicting values using Artificial Neural Networks.
Rock_Paper_Scissors_Spock	Rock Paper Scissors with “Lizard Spock” rules.
RSS_Manager	Reads and manages RSS news feeds.
Screenpet	A cute digital pet for your desktop.
Shape_Recognition	Identifies geometric shapes in images.
simple-chatbot	A basic rule-based conversational AI.

Project Name	Description
Simple_dice	A minimalist dice rolling tool.
Simple_Http_Server	A basic web server written in Python.
Simple_Image_resize	Resizes images to specific dimensions.
Slideshare to PDF	Downloads Slideshare presentations as PDFs.
Smart_Calculator	A calculator with advanced functionality.
Snake_game	The classic arcade snake game.
Socket_example	Demonstrates network communication via sockets.
spam_bot	Automates repetitive messaging (for educational use).
Speaking_Dictionary	A dictionary that reads definitions aloud.
Speaking_Wikipedia	Fetches and reads Wikipedia articles aloud.
Speech_To_Text	Converts spoken language into written text.
Spinning Donut	A 3D-rendering of a donut in the terminal.
Sqlite-crud	Simple database operations using SQLite.
Stack_structure	Implementation of the Stack data structure.
Star_Pyramid	Generates pyramid patterns using stars (*).
string_manipulator	Tool for common string transformation tasks.
Sudoku_solver	An algorithm to solve Sudoku puzzles.
TestTypingSpeed	Measures how fast you can type accurately.
Text_Predication	Basic predictive text algorithm.
TextEncryptor	Simple tools for encrypting and decrypting text.
TextToSpeech	Converts written text into spoken audio.
TEXTVENTURE	A text-based adventure game.
Tic_Tac_Toe	The classic Xs and Os game.
Todo_App	A productivity tool for managing tasks.
Tower-of_Hanoi	Solver for the Tower of Hanoi puzzle.
Triangle Calculator	Calculates properties of triangles.
Url_Shortener	Creates short aliases for long URLs.
web scraping- Find python jobs from a website	Scrapes job boards for Python roles.
Web scraping for book names	Extracts book titles from online stores.
Web Scraping iPhone from Flipkart	Specifically scrapes iPhone data from Flipkart.
website-builder	Simple tool to generate HTML websites.
Weights_on_different_planets	Calculates your weight across the solar system.
Whatsapp_Bot	Basic automation for WhatsApp.

Project Name	Description
Windows_Shutdown	Tool to schedule or trigger system shutdowns.
Word_Jumble	A word scrambling and unscrambling game.
Wordle_Aid	Helps you solve Wordle puzzles.
xls_to_xlsx	Converts old Excel formats to modern ones.
Youtube_video_download	Downloads videos or audio from YouTube.
Zombie_Game	A survival-themed game project.

... and many more!

3 Summary

In summary, this collection provides over 100 mini-projects to help you learn Python. From simple games like “Henman” and “Snake” to more advanced tools like “Address Validator” and “Face Recognition”, there is plenty to explore and learn from.

We hope you find these projects useful for your learning journey!

References